

Ag Testing - Consulting

Account No. : 602314

Soil Analysis Report

LOBATO, MICHAEL
1066 19 1/2 RD
FRUITA

CO 81521

Invoice No. : 1434509
Date Received : 11/05/2024
Date Reported : 11/07/2024

Results For : CSU WCRC FRUITA
Location : N-NOVEL BIOCHAR INJECTION TRIAL

Lab No. : 181088
ID : STRIP 1

Soil Health Assessment

Depth : 0 - 12

1:1 Soil pH	7.8	Soil Tests	
1:1 Soluble Salts, mmho/cm	0.53	Phosphorus Olsen P, ppm P	12.5
Excess Lime Rating	HIGH	Potassium NH4OAc, ppm K	151
Organic Matter, %LOI	2.3	Zinc DTPA, ppm Zn	1.46
		Iron DTPA, ppm Fe	6.4
Soil Respiration CO₂-C, ppm C	12.0	Manganese DTPA, ppm Mn	0.9
Water Extract		Copper DTPA, ppm Cu	0.68
Total Nitrogen, ppm N	11.1	Sulfate M-3, ppm S	78.1
Organic Nitrogen, ppm N	10.0	Calcium NH4OAc, ppm Ca	3535
Total Organic Carbon, ppm C	139	Magnesium NH4OAc, ppm Mg	382
Ammonium-N, H ₂ O ppm N	0.5	Sodium NH4OAc, ppm Na	300
Nitrate-N H ₂ O, ppm N	0.67		
Calculations		Soil Health	
Microbially Active Carbon (%MAC)	8.6	Soil Health Calculation	5.24
Organic C : Organic N	13.9	Cover Crop Suggestion	50% Legume 50% Grass
Organic Nitrogen Release, ppm N	3.4	Water Stable Aggregates (Mod) %	44
Organic Nitrogen Reserve, ppm N	6.6		Includes 0.25 -2.00 mm diameter sand grains

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Lab No. : 181088 Depth : 0 - 12
ID : STRIP 1

Phosphorus Olsen P, ppm P 12.5

Sum of Cations, me/100g	% Saturation				
	H	K	Ca	Mg	Na
22.5	0	2	78	14	6

Silica by 0.5 M Acetic Acid = 28.3

ppm Si. Ray

You can find a copy of our SHA

Guide at

<https://www.wardlab.com/resources/>

Recommendations

In Actual Pounds of Plant Nutrients per Acre

N Credit : 181088

Sub-Soils :

Crop	(Ward) Grass, T/A
Yield	2
Nitrogen N	35
Phosphorus P ₂ O ₅	15
Potassium K ₂ O	0
Sulfur S	0
Zinc Zn	0
Magnesium Mg	0
Iron Fe	0
Manganese Mn	6
Copper Cu	0

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Results For : CSU WCRC FRUITA
Location : N-NOVEL BIOCHAR INJECTION TRIAL

Lab No. : 181089
ID : STRIP 2

Soil Health Assessment

Depth : 0 - 12

1:1 Soil pH	7.9	Soil Tests	
1:1 Soluble Salts, mmho/cm	0.13	Phosphorus Olsen P, ppm P	11.2
Excess Lime Rating	HIGH	Potassium NH4OAc, ppm K	96
Organic Matter, %LOI	2.0	Zinc DTPA, ppm Zn	1.44
		Iron DTPA, ppm Fe	8.2
Soil Respiration CO ₂ -C, ppm C	16.0	Manganese DTPA, ppm Mn	1.0
Water Extract		Copper DTPA, ppm Cu	0.72
Total Nitrogen, ppm N	9.1	Sulfate M-3, ppm S	76.9
Organic Nitrogen, ppm N	8.2	Calcium NH4OAc, ppm Ca	3455
Total Organic Carbon, ppm C	126	Magnesium NH4OAc, ppm Mg	359
Ammonium-N, H ₂ O ppm N	0.5	Sodium NH4OAc, ppm Na	287
Nitrate-N H ₂ O, ppm N	0.41		
Calculations		Soil Health	
Microbially Active Carbon (%MAC)	12.7	Soil Health Calculation	5.27
Organic C : Organic N	15.5	Cover Crop Suggestion	50% Legume 50% Grass
Organic Nitrogen Release, ppm N	4.1	Water Stable Aggregates (Mod) %	44
Organic Nitrogen Reserve, ppm N	4.0		Includes 0.25 -2.00 mm diameter sand grains

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Location : N-NOVEL BIOCHAR INJECTION TRIAL

Lab No. : 181089 Depth : 0 - 12
ID : STRIP 2

Phosphorus Olsen P, ppm P 11.2

Sum of Cations, me/100g	% Saturation				
	H	K	Ca	Mg	Na
21.8	0	1	79	14	6

Silica by 0.5 M Acetic Acid = 26.2
ppm Si. Ray

Recommendations
In Actual Pounds of Plant Nutrients per Acre

N Credit : 181089

Sub-Soils :

Crop	(Ward) Grass, T/A
Yield	2
Nitrogen N	35
Phosphorus P ₂ O ₅	20
Potassium K ₂ O	25
Sulfur S	0
Zinc Zn	0
Magnesium Mg	0
Iron Fe	0
Manganese Mn	6
Copper Cu	0

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Results For : CSU WCRC FRUITA
Location : N-NOVEL BIOCHAR INJECTION TRIAL

Lab No. : 181090
ID : STRIP 3

Soil Health Assessment

Depth : 0 - 12

1:1 Soil pH	8.2	Soil Tests	
1:1 Soluble Salts, mmho/cm	0.07	Phosphorus Olsen P, ppm P	9.0
Excess Lime Rating	HIGH	Potassium NH4OAc, ppm K	75
Organic Matter, %LOI	1.8	Zinc DTPA, ppm Zn	1.12
		Iron DTPA, ppm Fe	6.6
Soil Respiration CO₂-C, ppm C	17.0	Manganese DTPA, ppm Mn	0.7
Water Extract		Copper DTPA, ppm Cu	0.61
Total Nitrogen, ppm N	8.5	Sulfate M-3, ppm S	113.3
Organic Nitrogen, ppm N	8.1	Calcium NH4OAc, ppm Ca	3060
Total Organic Carbon, ppm C	113	Magnesium NH4OAc, ppm Mg	323
Ammonium-N, H ₂ O ppm N	0.1	Sodium NH4OAc, ppm Na	287
Nitrate-N H ₂ O, ppm N	0.30		
Calculations		Soil Health	
Microbially Active Carbon (%MAC)	15.1	Soil Health Calculation	5.10
Organic C : Organic N	13.9	Cover Crop Suggestion	50% Legume 50% Grass
Organic Nitrogen Release, ppm N	4.9	Water Stable Aggregates (Mod) %	17
Organic Nitrogen Reserve, ppm N	3.2		Includes 0.25 -2.00 mm diameter sand grains

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Location : N-NOVEL BIOCHAR INJECTION TRIAL

Lab No. :	181090	Depth :	0 - 12			
ID :	STRIP 3					
Phosphorus Olsen P, ppm P						9.0
Sum of Cations, me/100g		% Saturation				
		H	K	Ca	Mg	Na
19.4		0	1	79	14	6
Silica by 0.5 M Acetic Acid = 21.8						
ppm Si. Ray						

Recommendations	
In Actual Pounds of Plant Nutrients per Acre	
N Credit :	181090
Sub-Soils :	
Crop	(Ward) Grass, T/A
Yield	2
Nitrogen N	35
Phosphorus P ₂ O ₅	25
Potassium K ₂ O	40
Sulfur S	0
Zinc Zn	0
Magnesium Mg	0
Iron Fe	0
Manganese Mn	7
Copper Cu	0

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Results For : CSU WCRC FRUITA
Location : N-NOVEL BIOCHAR INJECTION TRIAL

Lab No. : 181091
ID : STRIP 4

Soil Health Assessment

Depth : 0 - 12

1:1 Soil pH	8.2	Soil Tests	
1:1 Soluble Salts, mmho/cm	0.08	Phosphorus Olsen P, ppm P	9.9
Excess Lime Rating	HIGH	Potassium NH4OAc, ppm K	75
Organic Matter, %LOI	1.7	Zinc DTPA, ppm Zn	1.03
		Iron DTPA, ppm Fe	8.4
Soil Respiration CO₂-C, ppm C	13.0	Manganese DTPA, ppm Mn	1.1
Water Extract		Copper DTPA, ppm Cu	0.70
Total Nitrogen, ppm N	11.1	Sulfate M-3, ppm S	166.8
Organic Nitrogen, ppm N	10.8	Calcium NH4OAc, ppm Ca	3248
Total Organic Carbon, ppm C	123	Magnesium NH4OAc, ppm Mg	365
Ammonium-N, H ₂ O ppm N	0.1	Sodium NH4OAc, ppm Na	419
Nitrate-N H ₂ O, ppm N	0.22		
Calculations		Soil Health	
Microbially Active Carbon (%MAC)	10.6	Soil Health Calculation	5.10
Organic C : Organic N	11.3	Cover Crop Suggestion	50% Legume 50% Grass
Organic Nitrogen Release, ppm N	4.6	Water Stable Aggregates (Mod) %	16
Organic Nitrogen Reserve, ppm N	6.2	Includes 0.25 -2.00 mm diameter sand grains	

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Lab No. : 181091 Depth : 0 - 12
ID : STRIP 4

Phosphorus Olsen P, ppm P 9.9

Sum of Cations, me/100g	% Saturation				
	H	K	Ca	Mg	Na
21.3	0	1	76	14	9

The silica test is done. This sample has 132 ppm Si. Less than 6 ppm Si is low and 6-24 ppm Si is medium and greater than 24 ppm Si is high.
Ray

Recommendations

In Actual Pounds of Plant Nutrients per Acre

N Credit : 181091

Sub-Soils :

Crop	(Ward) Grass, T/A
Yield	2
Nitrogen N	35
Phosphorus P ₂ O ₅	20
Potassium K ₂ O	40
Sulfur S	0
Zinc Zn	0
Magnesium Mg	0
Iron Fe	0
Manganese Mn	6
Copper Cu	0